Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/30/2014
PI Name:	Rojas-Pierce, Marcela Ph.D.		
Project Title:	The Role of Vacuole Membrane Fusion in Plant Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	27695-7612	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/02/2013	**End Date:**	07/01/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AM49G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The long-term goal of this research is to identify and characterize mechanisms of plant vacuole biogenesis that are critical for gravity perception. A role of the vacuole in controlling gravity perception is well accepted, but the molecular mechanisms by which the vacuole interacts with amyloplasts or regulates their sedimentation during gravitropism are yet to be identified. We recently uncovered a novel vacuole phenotype in agravitropic mutant alleles of VTI11 in Arabidopsis. We predict that this vacuole phenotype of vti11 mutants has negative effects on the organization of actin cables or the cortical endoplasmic reticulum, and therefore prevent amyloplast sedimentation. In addition, we have uncovered a novel regulation of vacuolar membrane fusion by phosphoinositides. Based on our results, we developed the hypothesis that mechanisms of vacuole remodeling that are mediated by phosphoinositide signaling are critical for the perception or transduction of the gravitropic signal in plants. A unique assay for vacuole fusion that we developed will be used to determine the role of specific cellular components in membrane fusion and gravity perception. The		

	specific aims of this research project are to: 1) Determine subcellular phenotypes that impair gravity perception in itt3/vti11. 2) Identify the role of phosphoinositides in gravitropism. 3) Determine the role of the Adaptor Protein (AP-3) complex on vacuole biogenesis and gravitropism. Overall, this research will improve our understanding of vacuole biogenesis, the interactions between the vacuole and other cellular components involved in gravity perception, and the role of phosphoinosidides in gravitropism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute to the understanding of the molecular mechanisms that control the response of plants to gravity. In addition, the function of the vacuole in establishing the morphology and dynamics of other organelles will be characterized. These results will enhance the fundamental understanding how plant cells organize into a defined set of independent organelles.
Task Progress:	Personnel One graduate student Jiameng Zheng has been supported on this grant. She has completed the characterization of the new vti11 mutant allele itt3, which is now published. She also successfully advanced to candidacy in January 2014. She is currently setting up several constructs for subcellular localization of the VTI11 SNARE complex and analysis of protein-protein interactions with candidate genes. One postdoctoral scholar was recruited and hired in February 2014. Dr. Ashley Alvarez has been in charge of setting up the vertical stage and the proper conditions for imaging Arabidopsis seedlings during gravity stimulation. She has also made several constructs for shoot endodermis expression of some of the marker proteins that we would like to analyze. One undergraduate student, Samuel Lee, has conducted research for this project. He is involved in the analysis of vacuole morphology of several ap-3 mutants and their interaction with VTI11. Research progress The specific aims of this research project are to: 1) Determine subcellular phenotypes that impair gravity perception in itt3/vti11. 2) Identify the role of phosphoinositides in gravitropism. 3) Determine the role of the Adaptor Protein (AP-3) complex on vacuole biogenesis and gravitropism. We have made good progress on this project during the first year of the award. There was a delay on the hiring of a postdoc due to the time required to search, select, and hire a postdoc scholar. A new postdoc, Dr. Ashley Alvarez, was hired on February 1st. Nonetheless, significant progress has been made during the first year: a) We completed the submission of an article describing the vacuole phenotype of itt3 mutants and the possible role of vacuole fusion in gravity response. Jiameng Zheng, a graduate student supported on this grant since Fall 2013, is the first author in this publication. b) We have generated expression constructs for the visualization of subcellular markers in the shoot endodermis, and have started to introduce those into itt3 mutants. c) We have set up the vertical stage microscope for live-cell imaging of vacuoles and amyloplasts in wild type and in itt3 mutants during vacuole fusion experiments and during gravi-stimulation. This system allows us to image Arabidopsis seedlings in time-lapse experiments using a water objective with high numerical aperture. d) Started to identify the effects of vacuole fragmentation in itt3 on other subcellular organelles. e) We have acquired mutants for the AP-3 complex and initiated experiments to visualize their vacuole morphology. We have initiated crosses of several of these mutants to the vti11 alleles to determine if these two complexes genetically interact to control the morphology of the vacuole. Outreach: We hosted 15 students from a local Middle School in Wake County to expose them to research in plant biology. One of the activities that they did was comparing the vacuole morphology between wild type and itt3 mutants. Using these tools, we discussed the role of the vacuole in plant gravity response.
Bibliography Type:	Description: (Last Updated: 11/17/2022)
Articles in Peer-reviewed Journals	Zheng J, Han SW, Rodriguez-Welsh MF, Rojas-Pierce M. "Homotypic vacuole fusion requires VTI11 and is regulated by phosphoinositides." Molecular Plant. 2014 Jun;7(6):1026-40. Epub 2014 Feb 25. PubMed PMID: 24569132 ; http://dx.doi.org/10.1093/mp/ssu019 (NOTE: reported originally as "2014 Apr 3 [Epub ahead of print]" in April 2014.) , Jun-2014